

Daniele Uras

Biomedical Engineer — Machine Learning and Data Analysis

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Summary

Biomedical engineer with hands-on experience applying machine learning to real-world data, including physiological signals, clinical time series and medical images. Comfortable owning projects end to end, from problem definition and data preparation through implementation, validation and reporting of results. Available from November 2026.

Education

M.Sc. in Biomedical Engineering — Politecnico di Milano, Italy 09/2024 – 10/2026 (expected)
Bioinformatics track. Weighted average: 28.02/30.

B.Sc. in Biomedical Engineering — Università di Cagliari, Italy 09/2021 – 09/2024
Final grade: 105/110. Included a 5-month Erasmus research period at Łódź University of Technology, Poland.

Experience

Master's Thesis Researcher — Politecnico di Milano, Milan, Italy 11/2025 – 10/2026
Deep learning for EEG signal decoding

- Designed and trained a neural network architecture in Python and PyTorch that outperformed the previous best published result on a public EEG benchmark (BCI Competition 2020) using 30 times fewer parameters
- Designed and ran a systematic ablation study to isolate which architectural choices drove performance
- Identified a flaw in the dataset's standard preprocessing pipeline that degraded model performance, and quantified its impact

Bachelor Thesis Researcher — Łódź University of Technology, Poland 02/2024 – 07/2024
Data analysis for VR-induced postural stability (Erasmus programme)

- Designed the experimental protocol and analysed pedobarometric and centre-of-pressure data in MATLAB and Python, working in an international laboratory in English

Projects

- **ICU Mortality Prediction** (2025) — Clinical prediction pipeline on patient vital-sign data using XGBoost, 1D-CNN and SHAP, including feature engineering, patient-subgroup analysis and model interpretability. Python.
- **Narcolepsy Events Monitoring System** (2025) — Python GUI application for monitoring and predicting clinical events, with structured data workflows and visualisations. Python, SQL, XML.
- **EEG Sleep Stage Analysis** (2024) — Automated pipeline for sleep-stage classification from physiological signals, with feature extraction and validation. MATLAB, Python.
- **Brain Lesion Segmentation** (2024) — Medical image segmentation models on MRI volumes, evaluated with standard overlap metrics. MATLAB.

Skills

Programming and tools: Python, PyTorch, NumPy, pandas, scikit-learn, SQL, MATLAB, Git, LaTeX

Methods: Machine Learning, Deep Learning, Data Analysis, Signal Processing, Model Validation, Data Visualisation, Feature Engineering

Certifications

IELTS Academic — English, level C1 (CEFR)

Claude 101 — Anthropic

Languages

Italian: Native | **English:** C1 (IELTS Academic) | **Spanish:** A2, basic

Additional Activities

Elected Student Representative — Politecnico di Milano and Università di Cagliari 2022 – 2026
Elected for four years in total. Sat on departmental committees at the Department of Electronics, Information and Bioengineering, negotiating between students, faculty and administration on academic and organisational matters.

Member — Svoltastudenti, national student association 2022 – present
Focused on university governance.